RAINWISE V3.1 — Real-Time Flood & Multi-Threat Detection

Student: Hetvi Sheth

Department: Computer Science Engineering

Institute: Navrachana University

Abstract

This project focuses on real-time flood detection and multi-threat monitoring using deep learning and computer vision techniques.

The system combines SegFormer-B0 for flood segmentation and YOLO-World for zero-shot object detection including humans, vehicles, and crocodiles.

SigLIP was used for enhanced classification accuracy, achieving 98.89% detection performance.

LSTM forecasting models were integrated for 1-hour, 3-hour, and 6-hour flood prediction.

The model was optimized for edge deployment on Apple Silicon devices with real-time FPS performance.

Technologies Used

Technology Stack
PyTorch, SegFormer, YOLO-World, SigLIP, LSTM, OpenCV

Future Scope

This project can be extended with advanced AI models, cloud deployment, real-time scalability, and improved datasets for better accuracy and performance.

Conclusion

The project demonstrates practical implementation of AI, machine learning, or software technologies to solve real-world problems effectively.